Evaluating the Effectiveness of Promotional Email Campaigns on Customer Conversion:

A/B Testing Analysis Based on Hillstrom Email Marketing Dataset

Group D

Course: Statistical Reasoning for Business Decisions (1001)

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1 Business Needs Analysis

In this project, we were a project team from a consulting firm that was hired to provide data analysis services for a clothing retail company, who has been doing email marketing, hoping to bring customers back to the website and promote purchases by sending promotional emails. But management wasn't sure if these promotional emails were really effective.

Therefore, the client wants us to answer a realistic problem: Does promotional emails really increase customer purchase conversion (Chaffey & Ellis-Chadwick, 2022)?

For companies, promotional emails are a common marketing method, but sending emails itself is not the same as creating value. What they really concentrate about is whether emails will lead to more purchases. If emails don't increase conversions, then continuous delivery not only wastes marketing resources, but also reduces the user experience by overly intrusive customers (Ellis, 2020).

Therefore, the core decision-making issues of this project are:

Does sending promotional emails significantly increase customer conversion rates compared to not sending emails?

By comparing treatment groups and control groups, the impact of "whether you received the email" on customer behavior can be more clearly identified, helping business determines if email marketing is truly effective (Kohavi et al., 2020).

1.1 Primary Metric

The main metric for this project is the conversion rate, which is defined as:

$$\text{Conversion Rate} = \frac{\text{Number of Customers Purchased}}{\text{Total Customers}} \tag{1}$$

Conversion rate is chosen as a core metric because it is the most direct reflection of whether the campaign is delivering real business benefits (Montgomery, 2020).

In this analysis, we combined `Mens E-Mail` and `Womens E-Mail` into the `Any E-Mail` as the Treatment Group and compared them with the `No E-Mail`, defined as Control Group.

1.2 Supporting Metric

In addition to conversion rates, we also focus on visit rate as a secondary metric. Visit rates can help us determine whether an email is successful in bringing customers back to the website

(Sahni et al., 2018).

If the visit rate increases but the conversion rate does not increase significantly, it means that the email may have increased traffic but did not effectively convert into purchases; If both the visit rate and the conversion rate are up, it means that the email improves both the customer visit and the final purchase result.

1.3 Business Value

This analysis has three values for businesses.

1. It can help businesses determine whether promotional emails are worth continuing to run.
2. It can help businesses assess whether they need to further optimize their email content or delivery strategy.
3. It can provide a basis for subsequent more refined customer segmentation marketing.

Based on the results of our analysis, the purchase conversion rate for the `Any E-Mail` group is about 1.07%, and the `No E-Mail` group is about 0.57%, with an absolute lift of about 0.50 percentage points. At the same time, the visit rate of the mail group is higher than that of the control group. This result shows that promotional emails not only increase the probability of customers visiting your website, but may also lead to higher purchase conversions (Ansari & Mela, 2008).

2 Experiment Design and Effect Definition

2.1 Original Experiment Groups

Our original dataset consists of three groups:

1. **No Email (Control Group)**

   Customers in this group receive no promotional email at all during the experiment period. This serves as the baseline for our comparison.

2. **Men's Email**

   Customers in this group received a promotional email targeted at men, focusing on men's franchises and categories. This is one type of marketing treatment.

3. **Women's Email**

   Customers in this group received a promotional email targeted at women, focusing on women's franchises and categories. This is the second type of marketing treatment.

No Email is the control group, while Men's Email and Women's Email represent different types of promotional email treatments.

To clarify, Men's Email and Women's Email represent the type of promotional email sent to customers. That is our experimental treatment; it is not our customer segmentation variable. These are entirely different from the segment analysis, which describes pre-existing customer characteristics rather than the email treatment we applied. Moreover, the variables "men" and "women" in the data refer to customer gender attributes, while Men's Email and Women's Email indicate which email the customer received. These are separate concepts.

2.2 A/B Test Aggregation for Standard Analysis

In order to align the analysis with a standard A/B testing framework, this report combines the Men's Email and Women's Email groups into a single treatment group called Any Email. This aggregation allows the experiment to focus on the overall effect of receiving any promotional email compared with receiving none at all.

Group	**Definition**	**Explanation**
Control group	No E-Mail	Received no promotional email
Treatment group	Any E-Mail = Mens E-Mail + Womens E-Mail	Received either type of promotional email

This important aggregation clarifies and simplifies our core research question:

Does receiving any promotional email improve customer conversion compared to receiving no email at all?

2.3 Treatment Effect Definition: Uplift

The treatment effect is reflected by uplift, which equals the difference of the conversion rate of the treatment group minus the conversion rate of the control group.

$$\text{Uplift} = \text{Conversion Rate (Treatment Group)} - \text{Conversion Rate (Control Group)}$$

A positive uplift shows that the Any E-Mail has a higher conversion rate than the No E-Mail. However, we cannot be entirely sure unless the 95% confidence interval of the uplift is entirely above zero. In that case, the effect is considered statistically significant and the improvement is regarded as reliable.

3 Data Collection and Quality Check

The original dataset is collected from http://www.minethatdata.com/Kevin_Hillstrom_MineThatData_E-MailAnalytics_DataMiningChallenge_2008.03.20.csv

3.1 Data Dictionary

Variable Name	Explanation
Historical Customer Attributes	
recency	Months since the customer's last purchase (smaller value = more recently active)
history_segment	Group of annual consumption amount (not specific amount, but amount range)
history	Actual annual consumption amount
mens	Whether purchased men's clothing in the past year (1=Yes, 0=No)
womens	Whether purchased women's clothing in the past year (1=Yes, 0=No)
zip_code	Customer's region type
newbie	Whether is a new customer in the past 12 months (1=New, 0=Existing)
channel	Purchase channel used in the past year
Experiment Group Variable	
segment	AB test group (core treatment variable; not to be confused with "mens/womens")
Outcome Variables	
visit	Whether visited the website within 2 weeks after email sending (1=Yes, 0=No)
conversion	Whether made a purchase within 2 weeks after email sending (1=Yes, 0=No)
spend	Actual consumption amount within 2 weeks after email sending (0 if no purchase)

3.2 Data Quality Overview

See Appendix B.10.

The dataset contains 64,000 records, with one unique customer ID per row, eliminating duplicate observations and ensuring a valid sample frame. Original experiment design is in 3 groups Mens E-Mail, Womens E-Mail, No E-Mail. For the primary analysis, these are regrouped into 2 sets, the control group "No E-Mail", and combined treatment group "Any E-Maiil". For the data quality, the dataset shows a completeness, since no missing values are present across all variables, ensuring the data integrity. In addition, 3 original groups are evenly balanced ($\sim$33% each), confirming successful randomization. Overall, this dataset meets the quality standard and it could support a robust A/B test analysis (Kohavi et al., 2009).

4 Descriptive Statistics

4.1 Continuous Variables Summary

Variable	n	Mean	Median	SD	Min	Max
recency	64000	5.76	6.00	3.51	1.00	12.00
history	64000	242.09	158.11	256.16	29.99	3345.93
spend	64000	1.05	0.00	15.04	0.00	499.00

Recency demonstrates moderate customer activity frequency with a narrow value range, while history presents right-skewed distribution because of its much higher mean than median. Spend has a large value range, extremely low average value and zero median, indicating that most customers generate no consumption expenditure.

4.2 Binary Variables Summary

Variable	n_1 (Count of 1s)	pct_1 (%)
conversion	578	0.90
visit	9394	14.68
newbie	32144	50.22
mens	35266	55.10
womens	35182	54.97

channel	n	percentage (%)
Multichannel	7762	12.13
Phone	28021	43.78
Web	28217	44.09

Conversion rate is 0.90%, reflecting low purchasing transformation efficiency. Half of the samples are new users, and only 14.68% of customers have active visit behaviors. It is apparent that Web and Phone channels dominate customer access ways, jointly accounting for over 87%

of total traffic.

4.3 Categorical Variables Distribution

Zip Code Distribution

zip_code	n	percentage (%)
Rural	9563	14.94
Suburban	28776	44.96
Urban	25661	40.10

In this distribution, Suburban areas constitute the largest user group, followed by urban users, and rural residents make up the least share. Thus, this sample concentrates more on non-rural residential areas.

Original Segment Distribution

segment	n	percentage (%)
No E-Mail	21306	33.29
Mens E-Mail	21307	33.29
Womens E-Mail	21387	33.42

Three original groups' numbers show a balanced division.

5 Statistical Analysis and Hypothesis Testing

When p-value refers to 0, it means that the p-value is extremely low and is rounded to 0 in the output.

5.1 Main A/B Test: Any E-Mail vs No E-Mail

Since the business question is whether email drives the purchases. Therefore, conversion rate is selected as the primary KPI. Meanwhile, conversion is the binary outcome. In this way, the two-proportion test is utilized to compare the conversion rates between the Any E-Mail group and the No E-Mail group. In this section, the hypotheses is:

$$H_0 : p_{email} = p_{control} \quad H_1 : p_{email} \neq p_{control}$$

where p_{email} refers to the conversion rate of the Any E-Mail group and $p_{control}$ refers to the conversion rate of the No E-Mail group. The null hypothesis states that there is no difference in conversion rates between the two groups, while the alternative hypothesis states that there are different.

In R, we utilize following function:

```
prop.test(
  x = c(main_test$treatment_success, main_test$control_success),
  n = c(main_test$treatment_n, main_test$control_n),
  correct = FALSE
)
```

The result can be obtained:

Table 1: Two-Sample Test for Equality of Proportions without Continuity Correction

Test Result		**Interval and Estimates**	
X-squared	38.985	95% CI lower	0.003548427
df	1	95% CI upper	0.006360716
p-value	4.272e-10	sample prop 1	0.010680658
alternative hypothesis	two.sided	sample prop 2	0.005726087

The p-value is 4.272e-10 which is lower than 0.05 which refers to there is a statistically significant difference in conversion rates between the Any E-Mail group and the No E-Mail group. Meanwhile, the absolute lift is the difference of prop 1 and prop 2 which is 0.004955. In business terms, it represents that the promotional email could significantly improve the purchase conversion rate (Kohavi et al., 2020).

5.2 Secondary Outcome: Visit Rate

The secondary KPI is visit rate since we usually purchase the product after visiting the website. Therefore, it is also important to check whether the email could drive the visit. Similar to the main A/B test, the two-proportion test is utilized to compare the visit rates between the Any E-Mail group and the No E-Mail group. The hypotheses is:

$$H_0 : p_{visit,email} = p_{visit,control} \quad H_1 : p_{visit,email} \neq p_{visit,control}$$

where $p_{visit,email}$ refers to the visit rate of the Any E-Mail group and $p_{visit,control}$ refers to the visit rate of the No E-Mail group. The null hypothesis states that there is no difference in visit rates between the two groups, while the alternative hypothesis states that there are different.

In R, we utilize following function:

```
prop.test(
  x = c(visit_test$treatment_success, visit_test$control_success),
  n = c(visit_test$treatment_n, visit_test$control_n),
  correct = FALSE
)
```

The result can be obtained:

Table 2: Secondary A/B Test Result: Visit

Group	n	Success	Rate
Any E-Mail	42694	7132	0.167049
No E-Mail	21306	2262	0.106167

Metric	Value
Uplift	0.060882
CI low	0.055439
CI high	0.066325
z value	21.92178
p-value	0

The p-value is extremely low which represents as 0 and is lower than 0.05 which refers to there is a statistically significant difference in visit rates between the Any E-Mail group and the No E-Mail group. Meanwhile, the absolute lift is 0.060882 which represents that the promotional email could significantly improve the visit rate and could bring more traffic to the funnel (Manchanda et al., 2021).

5.3 Pairwise Comparison among Original Groups

The previous main test is to combine the Men's E-Mail group and the Women's E-Mail group together to compare with the No E-Mail group. However, it is also important to check whether there are differences between the Men's E-Mail group and the Women's E-Mail group. Therefore, pairwise comparison is adopted among the three groups. The hypotheses for each pairwise

comparison is:

$$H_0 : p_A = p_B \quad H_1 : p_A \neq p_B$$

In this section, the multiple test would conduct a type I error issue. Hence, we adopt the Holm-adjusted p-value to control the family-wise error rate (Holm, 1979). The steps of Holm's method are as follows:

1. Sort the raw p-values from the pairwise tests in ascending order.

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}, \quad m = 3$$

2. For each p-value, calculate the adjusted p-value using the formula:

$$p_{adj(i)} = \min\left(1, \max_{j \leq i}\left\{(m - j + 1) \cdot p_{(j)}\right\}\right)$$

3. Compare the adjusted p-values to the significance level to determine which comparisons are statistically significant. If $p_{adj(i)} < \alpha$, reject the null hypothesis for that comparison; otherwise, fail to reject the null hypothesis.

The result can be obtained:

Table 3: Pairwise Tests with Holm-Adjusted p-values

Comparison	Treatment Rate	Control Rate	Uplift	CI Low	CI High	p-value	Holm Adjusted p
Men's E-Mail vs No E-Mail	0.012531	0.005726	0.006805	0.005000	0.008610	0.000000	0.000000
Women's E-Mail vs No E-Mail	0.008837	0.005726	0.003111	0.001499	0.004723	0.000156	0.000312
Men's E-Mail vs Women's E-Mail	0.012531	0.008837	0.003694	0.001743	0.005644	0.000206	0.000312

The Holm-adjusted p-values for all three pairwise comparisons are less than 0.05, which refers to there are statistically significant differences in conversion rates between each pair of groups. In this way, the email works overall and Mens E-Mail group performs the best among the three groups (Thomas & Sullivan, 2022).

6 Deep Analysis and Insight Discovery

This section is an advanced part of AB test, aiming to present specific insights and useful suggestions for indicating effective segment, checking and controlling confounding variables' effects, giving structural analysis on uncovered patterns, robustly testing sample size reasonability.

6.1 Segmentation Analysis

The purpose of segmentation analysis is not to repeat the main hypothesis test, but to answer the question: Which customers is more effective for email? And for detailed detection, segmentation analysis can check email effect separately, for motivating marketing decisions by presenting insights (Liu & Lee, 2022).This report conducts segmentation by customer related variables including `channel`, `history_segment`, `newbie`, and `zip_code`. For each segment, we compute the No E-Mail conversion rate, Any E-Mail conversion rate, and the corresponding uplift, and present the required visualization. (See Appendix for figures.)

6.1.1 Channel Analysis

Table 4: Conversion Rate and Uplift by Channel

Segment	No E-Mail n	Any E-Mail n	No E-Mail rate	Any E-Mail rate	Uplift
Multichannel	2,606	5,156	0.69%	1.55%	0.86 pp
Phone	9,327	18,694	0.54%	0.89%	0.36 pp
Web	9,373	18,844	0.58%	1.11%	0.53 pp

From the channel segmentation, Multichannel customers show the highest uplift, suggesting that customers who have purchased through multiple channels in the past are more likely to be reactivated by email campaigns. These customers have richer brand and channel engagement histories, making promotional emails more effective at driving return visits and purchases (Verhoef et al., 2015).

See Appendix A.1.

6.1.2 History Segment Analysis

Table 5: Conversion Rate and Uplift by History Segment

Segment	No E-Mail n	Any E-Mail n	No E-Mail rate	Any E-Mail rate	Uplift
1) $0 - $100	7,612	15,358	0.42%	0.86%	0.44 pp
2) $100 - $200	4,836	9,418	0.41%	0.84%	0.43 pp
3) $200 - $350	4,044	8,245	0.69%	1.08%	0.39 pp
4) $350 - $500	2,124	4,285	1.13%	1.45%	0.32 pp
5) $500 - $750	1,652	3,259	0.54%	1.47%	0.93 pp
6) $750 - $1,000	622	1,237	0.48%	1.94%	1.46 pp
7) $1,000 +	416	892	1.44%	2.47%	1.02 pp

Analysis by `history_segment` reveals that customers with higher historical spending exhibit greater uplift. In particular, high-value segments such as $750-$1,000 and $1,000+ show substantially larger absolute improvements in conversion rate from email campaigns. This suggests that historical spending amount can serve as an important criterion for prioritizing email targeting (Gupta et al., 2006).

See Appendix A.2.

6.1.3 Newbie Analysis

Table 6: Conversion Rate and Uplift by Newbie Status

Segment	No E-Mail n	Any E-Mail n	No E-Mail rate	Any E-Mail rate	Uplift
0 (Existing)	10,611	21,245	0.77%	1.13%	0.36 pp
1 (Newbie)	10,695	21,449	0.37%	1.01%	0.63 pp

The `newbie` is a binary variable showing whether the experimented one is new customer over the past one year. This table shows that new customers have higher uplift than existing customers, indicating that email campaigns may have customer activation value for new buyers (Lemon & Verhoef, 2022).

6.1.4 Zip Code Analysis

Table 7: Conversion Rate and Uplift by Zip Code

Segment	No E-Mail n	Any E-Mail n	No E-Mail rate	Any E-Mail rate	Uplift
Rural	3,139	6,424	0.83%	1.26%	0.43 pp
Suburban	9,625	19,151	0.50%	1.00%	0.50 pp
Urban	8,542	17,119	0.56%	1.07%	0.51 pp

For zip code segmentation representing customers' locations, the uplift differences among Urban, Suburban, and Rural areas are relatively small. Therefore, it is roughly reasonable that geographic type is not a key driver of heterogeneity in this dataset.

6.2 Confounding Variable Discussion

It is common that the result of the experiment may be affected by some confounding variables. This section follow it to identify potential confounders using the change-in-estimate method with logistic regression (Gelman & Hill, 2006). The procedure is as follows:

6.2.1 Formula and Result

Crude Model (without confounders):

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_{\text{treat}} \cdot \text{treat}$$

$$\text{OR}_{\text{crude}} = e^{\hat{\beta}_{\text{treat}}}$$

Adjusted Model (with candidate confounder Z):

$$\log\left(\frac{p}{1-p}\right) = \beta_0' + \beta_{\text{treat}}' \cdot \text{treat} + \gamma \cdot Z$$

$$\text{OR}_{\text{adjusted}} = e^{\hat{\beta}_{\text{treat}}'}$$

Change-in-Estimate Criterion:

$$\Delta = \frac{|\text{OR}_{\text{adjusted}} - \text{OR}_{\text{crude}}|}{\text{OR}_{\text{crude}}} \times 100\%$$

Seeing those independent variables effect on conversion, if $\Delta \geq 10\%$, the variable Z is considered a potential confounder. Candidate variables include only pre-experiment customer characteristics: `recency`, `history`, `zip_code`, `newbie`, `channel`, `mens`, and `womens`. In this confounding variable test, `visit` and `spend` are excluded from confounding variable assessment because they occur after the email treatment (post-treatment variables), considering that they would undermine causal interpretation.

Table 8: Confounding Variable Identification Based on Odds Ratio Change

Variable	Crude OR	Adjusted OR	OR Change (%)	Judgment
recency	1.8746	1.8777	0.16%	Not a confounder (change $<$ 10%)
history	1.8746	1.8697	0.26%	Not a confounder (change $<$ 10%)
zip_code	1.8746	1.8728	0.09%	Not a confounder (change $<$ 10%)
newbie	1.8746	1.8749	0.02%	Not a confounder (change $<$ 10%)
channel	1.8746	1.8760	0.07%	Not a confounder (change $<$ 10%)
mens	1.8746	1.8749	0.02%	Not a confounder (change $<$ 10%)
womens	1.8746	1.8732	0.07%	Not a confounder (change $<$ 10%)

See Appendix A.3.

The results show that all candidate variables produce treatment OR changes well below 10%, with the maximum change being only 0.26%. Therefore, according to the change-in-estimate criterion, none of these variables are identified as major confounders. This finding aligns with the logic of randomized experimentation: random assignment typically reduces the confounding influence of pre-experiment customer characteristics on the treatment effect (Imbens & Rubin, 2015).

6.2.2 Logistic Regression Overview

We conducted a full logistic regression model adjusting for all pre-treatment customer characteristics, including all related variable.

$$\begin{aligned}\log\left(\frac{p}{1-p}\right) = \beta_0 &+ \beta_1 \cdot \text{treat} + \beta_2 \cdot \text{recency} + \beta_3 \cdot \text{history} \\ &+ \beta_4 \cdot \text{zip_code}_{\text{Suburban}} + \beta_5 \cdot \text{zip_code}_{\text{Urban}} + \beta_6 \cdot \text{newbie} \\ &+ \beta_7 \cdot \text{channel}_{\text{Phone}} + \beta_8 \cdot \text{channel}_{\text{Web}} + \beta_9 \cdot \text{mens} + \beta_{10} \cdot \text{womens}\end{aligned}$$

where:

- $p = P(\text{conversion} = 1 \mid X)$

See Appendix 9.

After controlling for these covariates, the treatment (email) still remained strongly and significantly associated with conversion with the remaining figure of odds ratio which is 1.87, reinforce our Hypothesis Testing (Hosmer et al., 2013).

6.3 Pattern Discovery

Pattern discovery is a section for finding out the indicators or insights for this AB test, aiming to synthesize key business-relevant patterns identified from preceding analysis. (See Appendix for figures.)

6.3.1 Uplift Effect by Mens' E-Mail

First, promotional emails in general significantly improve customer conversion, while the Mens' E-Mail treatment shows a stronger effect than the Womens' E-Mail treatment, indicating that different email contents should not be simply regarded as identical treatments. This heterogeneity in message effectiveness implies that content design and targeting strategy deserve further differentiation in future campaigns (Bleier et al., 2020).

Table 9: Conversion Rates by Treatment and Control Groups

Comparison	Treatment Rate	Control Rate
Mens E-Mail vs No E-Mail	0.01253	0.00577
Womens E-Mail vs No E-Mail	0.00884	0.00577
Mens E-Mail vs Womens E-Mail	0.01253	0.00884

6.3.2 Lift by Visit and Group

Second, emails generate a much larger lift in visit rate than in conversion rate. This pattern confirms that promotional emails are effective at driving customers back to the website, but successful purchase conversion still requires strong support from, for instance, landing-page design, promotional offers, and personalized product recommendation strategies. The gap between visit lift and conversion lift reveals clear opportunities to optimize the post-click conversion path (Lambrecht & Tucker, 2019).

See Appendix A.4.

6.3.3 Uplift by Channels and History Consumption

Third, the effectiveness of email interventions varies noticeably across customer segments. Multichannel customers and those with high historical spending exhibit stronger uplift and thus represent higher-priority targets for email marketing.

See Appendix A.5.

The uplift heatmap supports simultaneous visualization of treatment effects across two dimensions. Overall, segments associated with high historical spending and Multichannel/Web channels show particularly strong positive uplift. Cells labeled as blank or "No data" generally indicate insufficient sample size to form a reliable subgroup. It complements the heatmap by displaying not only the magnitude of uplift but also statistical uncertainty. Some subgroups appear to have large uplift values, but these estimates may be unreliable due to small sample sizes and wide confidence intervals. Therefore, business decisions should not rely solely on nominal uplift size; they must also consider sample size and the stability of confidence intervals (Rozenblit & Koralov, 2023).

See Appendix A.6.

The forest plot demonstrates specific patterns by four key element variables including history segments, channel, newbie and zip-codes, finalizing specific group with detailed categories which can maximize the conversion effect. This can help decision makers for targeting customers for gaining bigger conversion and profit. And this plot shows that customers consuming "more than $ 750", "receiving emails by multichannel such as phone and website", "usually

in Urban area", "just join this fashion shop in the previous 1 year", tends to be increase more conversion uplift.

6.4 Power Analysis

Power analysis is used to evaluate whether the experimental sample size is sufficient to detect our meaningful differences in conversion rates (Cohen, 1988). Since the Any E-Mail group is a combination of two email arms and is approximately twice as large as the No E-Mail control group, this report uses an approximation of the two-proportion test with unequal sample sizes. The test statistic for comparing two proportions is defined as:

$$z = \frac{\hat{p}_1 - \hat{p}_0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_0} + \frac{1}{n_1}\right)}}$$

where $\hat{p}_0$ and $\hat{p}_1$ denote the conversion rates of the control and treatment groups, $\hat{p}$ is the pooled proportion, and n_0, n_1 represent the sample sizes of the two groups.

For the observed uplift of 0.495 percentage points, the estimated power of the current sample is approximately 1.000, meaning the experiment has extremely high ability to detect the measured effect. According to smaller MDE needing bigger sample size, even when the minimum detectable effect (MDE) is set to 0.25 percentage points, the sample with power 0.947 still achieves ideal statistical power above the conventional threshold of 0.80. Together, they provide robust evidence for the reliability and practical relevance of the estimated email treatment effect.

See Appendix B.12.

7 Recommendations and Limitations

7.1 Result Interpretation

The results suggest that promotional emails have a positive short-term effect on customer conversion. Compared with the No E-Mail group, the Any E-Mail group show a higher conversion rate, indicating that promotional emails can help activate customer and encourage some of them to make purchases within the observed period. From a business perspective, this means that email marketing can be used as an effective short-term activation tool, especially when

the company aims to bring customers back to the brand and increase immediate conversion opportunity (Ansari & Mela, 2008).

However, the results also show that the visit lift are larger than the conversion lift. Promotional emails are more effective at driving customers back to the website than directly converting them into buyers. This means that emails can successfully generate customer attention and website traffic, but the final purchase decision still depend on the post-click experience. Therefore, the business value of email campaigns should not be evaluated only by whether emails are sent successfully or whether customers click the email. The company also needs to consider whether the landing page design, promotional offer, product recommendation, and checkout process can support the final conversion (Lambrecht & Tucker, 2019).

The segmentation results further show that the email effect are not the same across all customer groups. Multichannel customers shows the highest uplift among channel segments, suggesting that customers with richer brand and channel engagement history are more likely to be reactivated by promotional emails (Verhoef et al., 2015). In the channel analysis, Multichannel customers have an uplift of 0.86 percentage points, which is higher than Web and Phone customers. This indicates that customers who have interacted with the brand through multiple channels may have stronger familiarity with the brand and higher response potential.

Historical spending are also an important source of heterogeneity. The analysis by history segment show that customers with higher historical spending generally exhibit stronger uplift. In particular, the \$750-\$1,000 segment and the \$1,000+ segment show relatively large improvement in conversion rate after receiving emails. This suggests that historical spending can serve as an important criterion for prioritizing email targeting (Gupta et al., 2006). From a business perspective, high historical spending customers may already have stronger purchase intentions, brand familiarity, or product interest, so promotional emails may be more effective in reactivating them.

In addition, the Newbie segment also show meaningful activation potential. Newbie customers has a higher uplift than existing customers, which indicates that email campaigns may be useful for activating customer who joined or purchased recently (Lemon & Verhoef, 2022). This suggests that email marketing should not only focus on existing high-value customers, but can

also be used to build early engagement with newly acquired customers.

Overall, the results indicates that promotional emails can improve short-term customer conversion, but their effect is stronger for some customer segments than for others. Therefore, the key business interpretation are that email campaigns should not be treated as a one-size-fits-all strategy. Instead, the company should use customer segmentation to identify groups with higher response potential and design more targeted email strategy (Thomas & Sullivan, 2022).

7.2 Limitations

Although the analysis provide useful evidence for the effectiveness of promotional emails, there is still three limitations that should be considered when interpreting the result.

First, the analysis may suffer from time extrapolation bias. The current data only observe customer behavior within two weeks after the email campaign. Therefore, the results mainly reflect short-term outcomes, such as short-term visit and short-term conversion. However, short-term conversion may not fully represent long-term repeat purchase, customer retention, unsubscribe behavior, or customer lifetime value. If the two-week result are directly extended to long-term business performance, the conclusion may overstate or misrepresent the long-term marketing effect. Therefore, the current findings should be interpreted as evidence of short-term campaign effectiveness rather than as proof of long-term customer value improvement.

Second, the analysis may also involve survivorship bias. If some customers unsubscribes, become inactive, or stop interacting with the brand after receiving promotional emails, they may be less visible in later observations. In that case, the analysis may focus more on customers who remain active or observable, while customers with negative reaction may be underrepresented. This could lead to an overly positive interpretation of the email campaign, because long-term negative responses, such as unsubscribe behavior or reduced engagement, are not fully captured. Therefore, although the campaign show positive short-term effects, the company should still be cautious about assuming that all customer reactions are positive.

Third, there are a risk of heterogeneous treatment effect bias. In the main analysis, Mens E-Mail and Womens E-Mail is combined into the Any E-Mail group. This design makes the comparison between treatment and control clearer, because it allows the analysis to compare customers who received any promotional email with customers who received no email. How-

ever, this aggregation may hide differences between specific email contents. Mens E-Mail have a stronger conversion effect than Womens E-Mail, and the conversion rate of Mens E-Mail is higher than that of Womens E-Mail. This indicates that different email creatives should not be simply regarded as identical treatments.

Therefore, the average effect of Any E-Mail may not accurately represent the effect of each specific email type. If one type of email perform better than the other, the combined estimate may hide useful content-level insight. For business decision-making, this means that the company should not only ask whether sending emails are effective, but also ask which email content is more effective for which customer group (Bleier et al., 2020).

These limitations does not reject the positive email effect found in the analysis. Instead, they suggests that the results should be interpreted carefully. The findings are useful for short-term marketing decisions, but future analysis are still needed to evaluate long-term effects, customer retention, unsubscribe behavior, and differences across email creatives.

7.3 Future Work and Business Recommendations

Based on the result and limitation above, we recommend that the company improves future email campaigns from four main directions.

First, we recommend that the company prioritize high-uplift customer segments. The segmentation analysis show that Multichannel customers, high historical spending customers, and Newbie customers has stronger response potential. Therefore, the company should not send promotional emails to all customers in the same way. Instead, it should prioritize customers who are more likely to respond positively to emails. Specifically, Multichannel customers should be treated as a key target group because they shows the highest uplift among channel segments. High-spending customers should also receive more targeted email campaigns because the history segment analysis show that higher historical spending is associated with stronger uplift. Newbie customers should also be included in future targeting strategies because email campaigns may help activate them at an early stage.

Second, we recommend that the company optimizes the visit-to-conversion path. Since the results shows that emails generate a larger lift in visit rate than in conversion rate, the company should not only focus on increasing website visits. More importantly, it should improve the

customer experience after users click the email. Therefore, we recommend that the company improve landing page design, strengthen personalized product recommendations, make promotional offer clearer, and simplify the checkout and promotion redemption process. These action can help reduce customer drop-offs after the email click and improve the probability that website visits will become final purchases.

Third, we recommend that the company tests email contents separately in future experiments. Since Mens E-Mail and Womens E-Mail may have different effects, future campaigns should not only combine them into Any E-Mail. Instead, the company should separately test different email creatives and compare their performance directly. For example, the company can test different subject lines, discount levels, sending frequencies, product categories, and personalization strategies. This can help the company identify which type of email content work best for different customer segments. It can also help reduce heterogeneous treatment effect bias, because the company will no longer rely only on an average treatment effect that may hide content-level difference.

Fourth, we recommend that future research extends the observation window. The current two-week observation period are useful for evaluating short-term campaign effects, but it is not enough to understand long-term customer behavior. Therefore, the company should track customer outcomes over a longer period, such as one month or three months. In addition to short-term conversion, the company should also measures repeat purchase, customer retention, unsubscribe rate, and long-term customer value. This can help reduce time extrapolation bias and survivorship bias, because the company can observe whether customers continue to engage with the brand or leave after receiving promotional emails.

Overall, future campaign should combine targeted delivery, content-level testing, and long-term customer tracking. We recommend that the company first use segmentation result to identify high-response customers, then optimize the post-click conversion path, then run separate A/B test for different email contents, and finally evaluate long-term outcomes. This approach can help the company improve marketing efficiency, avoid a one-size-fits-all strategy, and maximize the business value of promotional email campaigns (Kohavi et al., 2020).

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A Appendix Figures

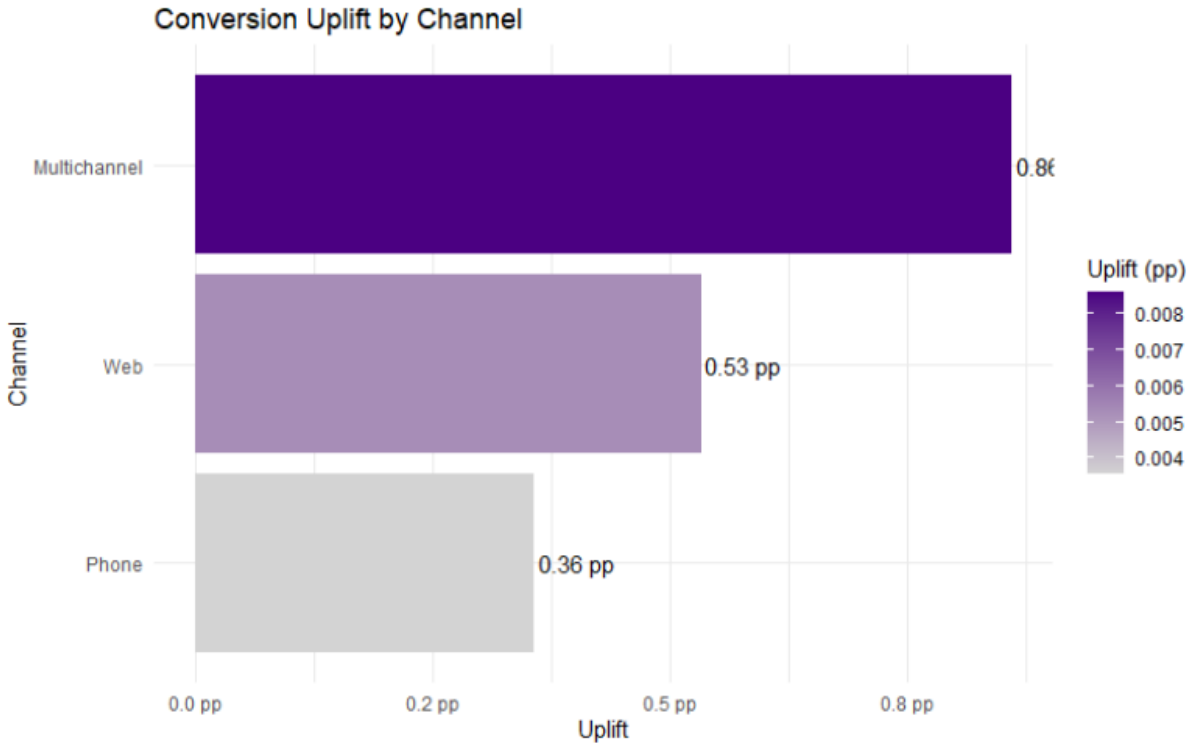


Figure A.1: Email Uplift by Channel

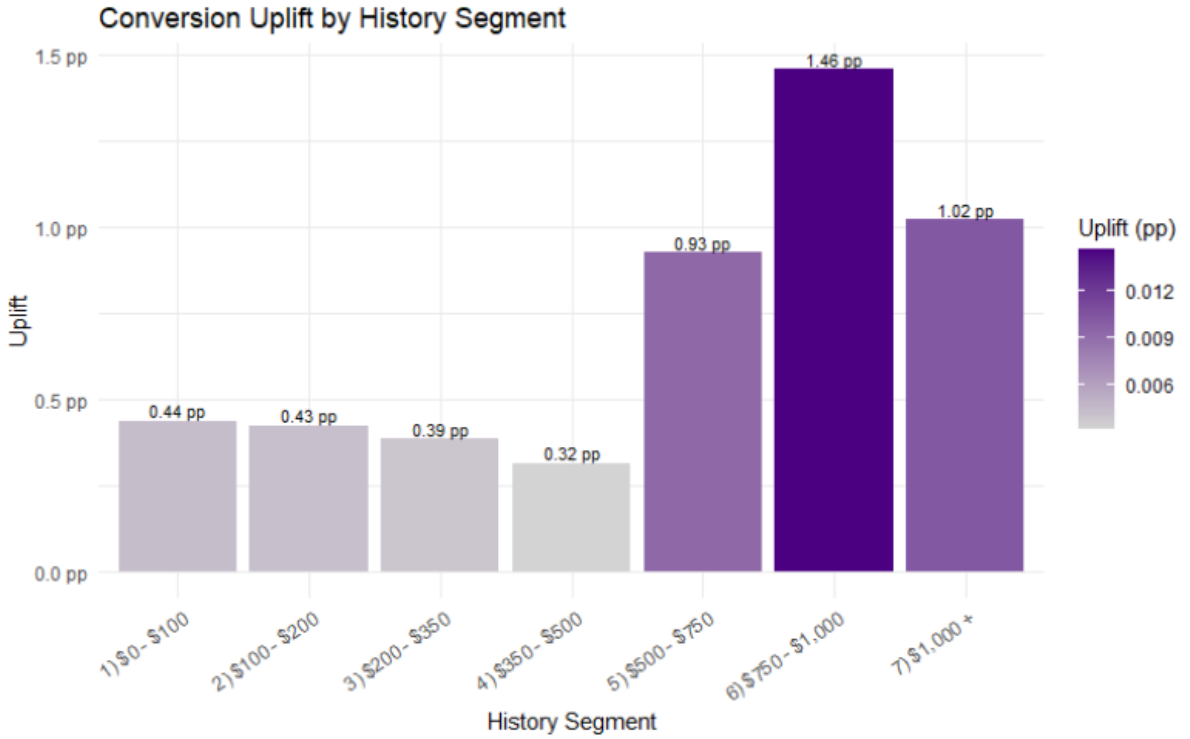


Figure A.2: Email Uplift by Historical Spending

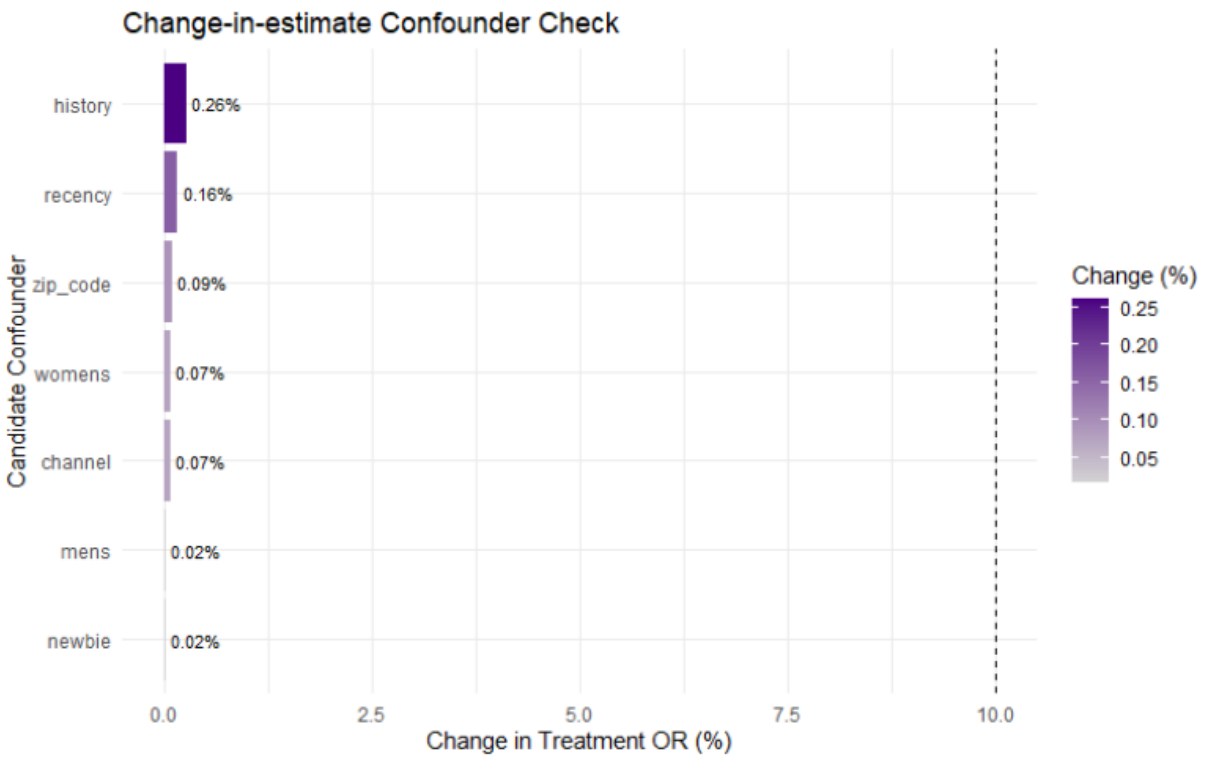


Figure A.3: Confounding Variable Check

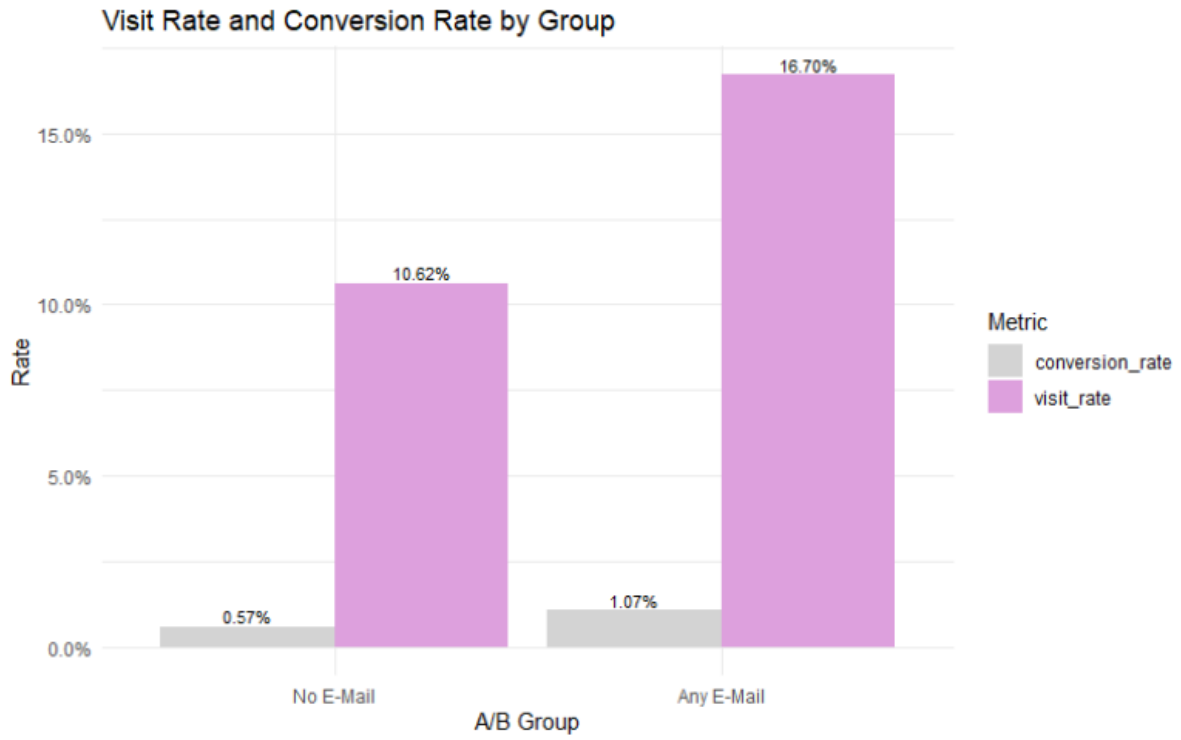


Figure A.4: Patterns of Uplift Across Segments and Visits

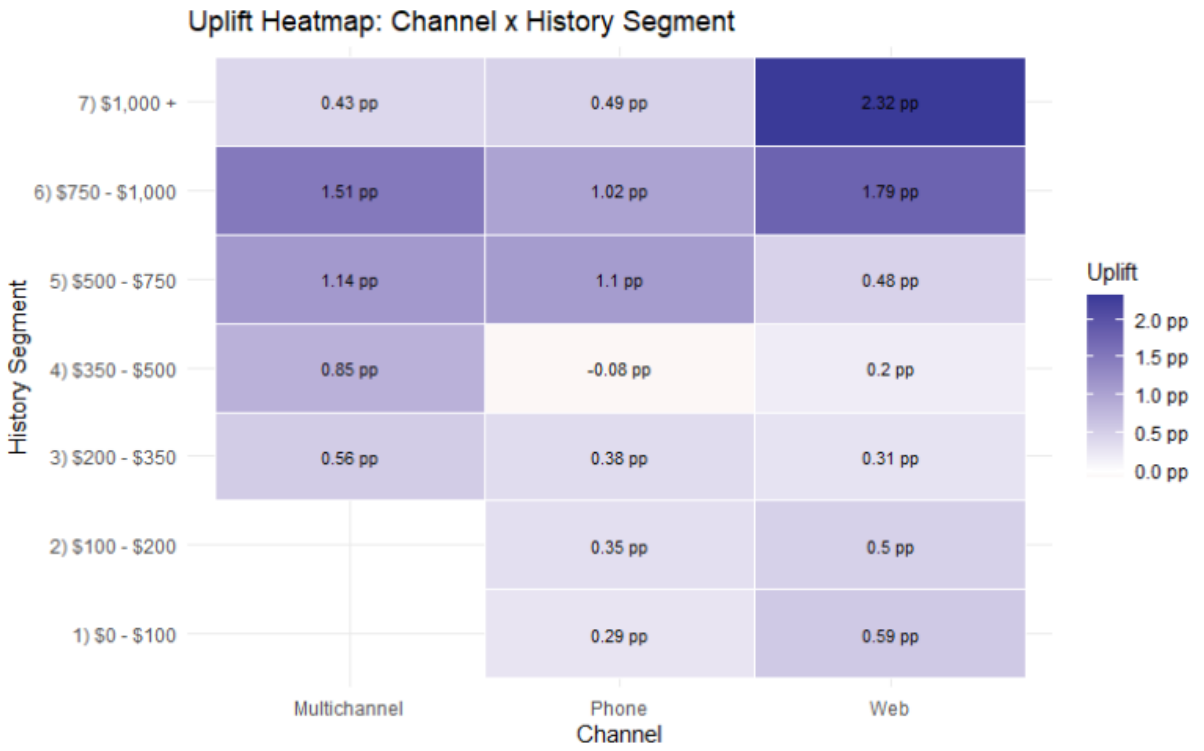


Figure A.5: Uplift Heat Map Across Channels and History Segments

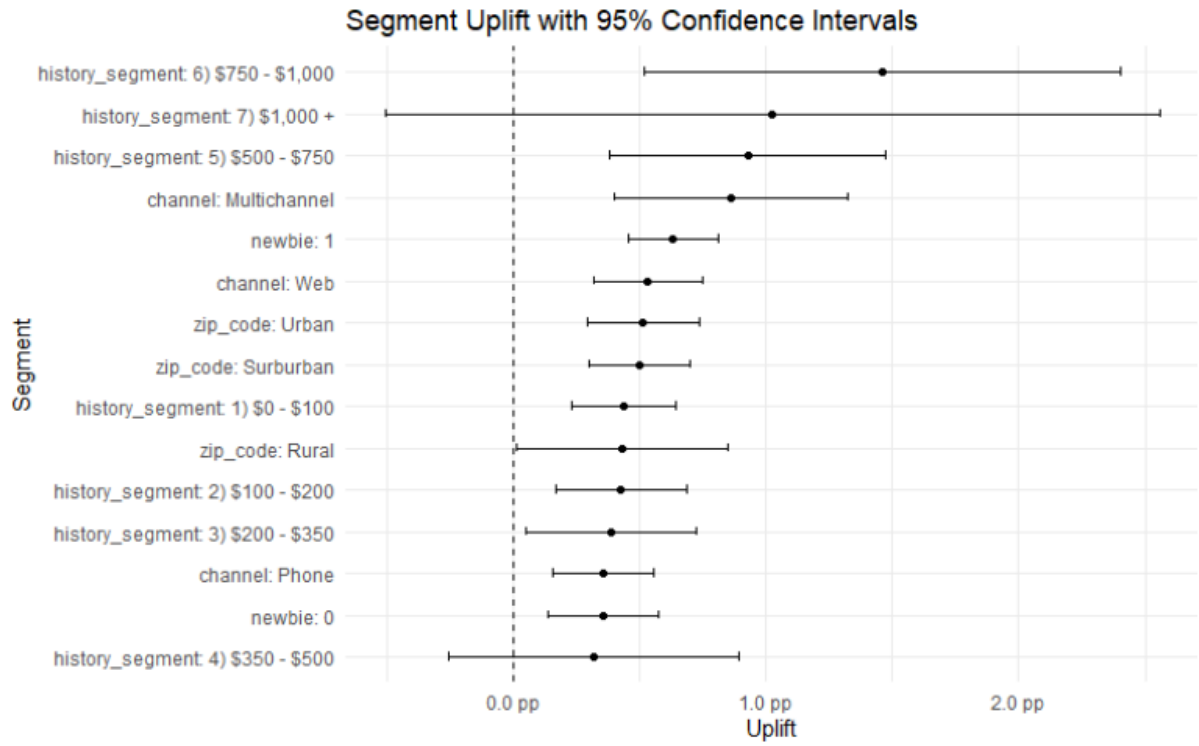


Figure A.6: Patterns' Effect Rank by Forest Plot

B Appendix Tables

Table B.10: Data Quality Overview

Item	Value
Total rows	64000
Unique customer IDs	64000
Original experiment groups	3 (Mens E-Mail, Womens E-Mail, No E-Mail)
Main AB groups	2 (Any E-Mail, No E-Mail)
Total missing values	0

Table B.11: Full Logistic Regression Odds Ratios

Term	Estimate	Std. Error	Odds Ratio	CI Low	CI High	P-value
(Intercept)	-5.0463	0.2326	0.0064	0.0041	0.0101	< 0.001
treat	0.6283	0.1024	1.8744	1.5336	2.2908	< 0.001
recency	-0.0593	0.0128	0.9424	0.9190	0.9664	< 0.001
history	0.0007	0.0002	1.0007	1.0004	1.0010	< 0.001
zip_codeSurburban	-0.2960	0.1171	0.7438	0.5913	0.9357	0.0115
zip_codeUrban	-0.2195	0.1177	0.8029	0.6375	1.0112	0.0621
newbie	-0.4316	0.0906	0.6494	0.5438	0.7756	< 0.001
channelPhone	-0.1737	0.1305	0.8406	0.6509	1.0855	0.1832
channelWeb	0.0087	0.1273	1.0088	0.7860	1.2946	0.9452
mens	0.3629	0.1266	1.4375	1.1217	1.8422	0.0041
womens	0.4928	0.1287	1.6369	1.2718	2.1067	0.0001

Table B.12: Power Analysis for Email Campaign (Control Rate = 0.573%)

Metric	Observed uplift	MDE 0.25 pp	MDE 0.50 pp	MDE 0.75 pp	MDE 1.00 pp
Control rate (%)	0.573	0.573	0.573	0.573	0.573
Treatment rate (%)	1.068	0.823	1.073	1.323	1.573
Uplift (pp)	0.495	0.25	0.50	0.75	1.00
Control n used	21,306	21,306	21,306	21,306	21,306
Treatment n used	42,694	42,694	42,694	42,694	42,694
Power	1.000	0.947	1.000	1.000	1.000
Required control n	–	13,340	3,985	2,058	1,318
Required treatment n	–	26,732	7,986	4,124	2,642